M55M1 Embedded AI Introduction

Max Dokukin

10/02/2025

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Microcontroller Platform

Feature: A Audio, C CAN, CF CAN FD, E Ethernet, L LCD, T Touch Key, U USB, V Video Codec, M Motor, P Power, N NPU, Low Power, TrustZone, AEC-Q100

Operating Voltage: 1.8 1.8V, 2.4 2.4V, 3.3 3.3V, 5 5V

Operating Frequency: 800 MHz, 650 MHz, 300 MHz, 220 MHz, 200 MHz, 120 MHz, 72 MHz, 64 MHz, 24 MHz, 7 MHz

8051

- ML56 (1.8, 3.3) T L
- ML54 (1.8, 3.3) L
- ML51 (1.8, 3.3, 5)
- MG51 (3.3, 5)
- MS51 (3.3, 5)
- N76E (3.3, 5)
- MUG51 (1.8, 3.3, 5)

Cortex®-M0

- NUC1261 (3.3, 5) U
- NUC126 (3.3, 5) U
- M091 (3.3)
- M031G (3.3)
- M029G/M030G (3.3)
- M032 (1.8, 3.3) U
- M031 (1.8, 3.3)
- NUC029 (3.3, 5) U
- NUC230 (3.3, 5) C
- NUC131U (3.3, 5) C
- NUC1311 (3.3, 5) C
- ISD9160 (3.3, 5) A
- ISD91500 (1.8, 3.3) A
- ISD91200 (1.8, 5) A
- N574 (1.8, 5) A T
- N572 (1.8, 5) U A T
- N570 (1.8, 5) A T
- M0A23 (3.3, 5) C
- M071 (3.3, 5)
- M051 (3.3, 5)
- Mini51 (3.3, 5)
- Nano100 (1.8, 3.3)

Cortex®-M23

- M2354 (1.8, 3.3) L U
- M2L31 (1.8, 3.3) T C U
- M2A23 (3.3, 5) CF
- M2351 (1.8, 3.3) U
- NUC1263 (3.3, 5) U
- NUC1262 (3.3, 5) U
- M263 (1.8, 3.3) C U
- M262 (1.8, 3.3) U
- M261 (1.8, 3.3)
- M258 (1.8, 3.3, 5) T L U
- M256 (1.8, 3.3, 5) T L
- M254 (1.8, 3.3, 5) L
- M253 (1.8, 3.3, 5) CF U
- M252 (1.8, 3.3, 5) U
- M251 (1.8, 3.3, 5)
- M2003 (3.3, 5)

Cortex®-M4

- ISD94100 (1.8, 3.3) A U
- M460 (1.8, 3.3) E CF U
- M480 (1.8, 3.3) E C U
- M433 (1.8, 3.3) C U
- M471 (3.3, 5) U
- KM1M4BF (3, 5) M
- M453 (3.3, 5) C
- M452 (3.3, 5) U
- M451 (3.3, 5)

Cortex®-M55

- M55M1 (1.8, 3.3) N E CF U
- M5531 (1.8, 3.3) E CF U

Cortex®-M7

- KM1M7AF (5) CF P
- KM1M7BF (5) M

MPU

Cortex®-A35

- MA35D1 V E CF L U
- MA35D0 E CF L U
- MA35H0 V E CF L U

Arm9™

- NUC980 E C U
- NUC970 E C L U
- N9H V E C L U
- N329 V E L U

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Product Information / Documents / Selection

- **Online Support**: NuForum / Sales Support Mailbox / Online Chat
- **Social Media & Knowledge Base**: LinkedIn / Facebook / Twitter / WeChat
- **Video Platform**: YouTube / bilibili
- **Open Resource**: Github / Gitlab / Gitee

Online buy
- Nuvoton Direct
- Tmall
- TechDesign
- DigiKey

Digital Platform

Development Platform

arm KEIL (+ VSCode) IAR Embedded Workbench NuEclipse
IDE

Nu-Link2-Pro/ Nu-Link2-Me
Nu-Link-Gang/ Gang Writer
Debugger & Programmer

NuMaker Series /
Motor Control Series / Power Control Series
Evaluation Board

MCU BSP (API Compatible) / Linux BSP/ Peripheral Driver / Library / Rich Sample Code
Third party resource
BSP & Example Code

NuIDE/ UltraoIO/ VR Lib / CodeGenerator/ PinView / PinConfig/ ClockConfig / LCDView / TouchView/ ICP/ISP Programming/ Nu-Link Command/ USB to Serial Port/ NuAudio
Software Tool(NuTool)

Platform

HMI Platform
AppWizard emWin SEGGER The Embedded Experts
Qt
SQUARELINE LVGL

IoT Device and Gateway
arm PELION amazon web services Alibaba Cloud Microsoft Azure Allxon
arm MBED AliOS Azure RTOS RT-Thread RTOS Linux OpenWrt
WiFi LTE NB-IoT LoRa Bluetooth

DALI 2 Solutions
Supports all control gear and control/ input device certified library protocol.

102	103
202	301
206	302
207	303
208	304
209	

Reference Design Platform

BMS	LCD Touch Key	Cold Chain	Thermostat
Lighting Control	HDMI2.1	USB Type-C PD3.1	AR & VR

8051 / M0 / M23 / M4 / M55 / M7 / Arm9 / A35 based Microcontroller

NuMicro® Microcontroller Platform

IoT Startup

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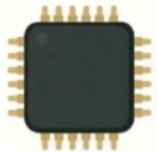
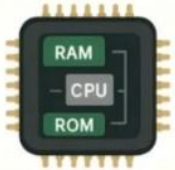
Difference Between Microprocessor and Microcontroller

Attribute	Microprocessor 	Microcontroller 
Core Composition	A CPU (central processing unit) on a single chip	A single chip with CPU, memory, and peripherals
Application Domain	Used in computers and high-end systems	Used in embedded systems and automation
Chip Integration	Only CPU (external RAM, ROM, I/O needed)	CPU + RAM + ROM + I/O + Timers on one chip
Power Consumption	High	Low
Cost	More expensive	Less expensive
Performance	Higher performance, used for complex tasks	Lower performance, used for specific control tasks
Typical Use Cases	Laptops, desktops, servers	Washing machines, microwave ovens, remote controls, IoT

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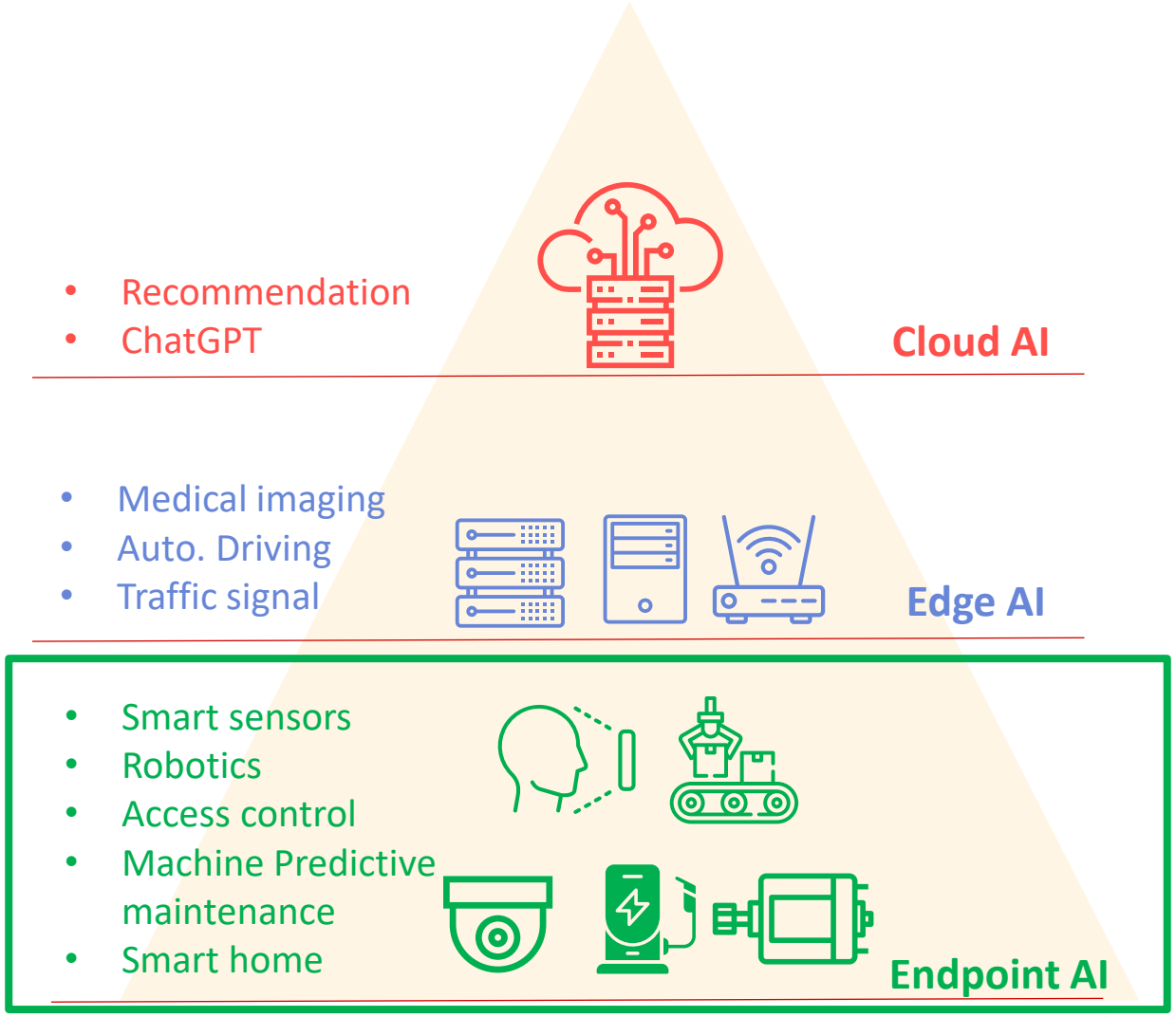
AI Architecture Layers

- **AI**
  - Simulation of human intelligence in machines
  - Cloud AI, Edge AI, Endpoint AI
- **Endpoint AI**
  - Deploying AI technology directly on endpoint devices to augment the intelligence of the system
- **Benefits of Endpoint AI**
  - Intelligence
  - Low latency
  - Privacy
  - Offline operation
  - Power Efficiency
  - Economy



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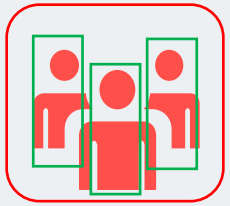
Endpoint AI's Applications

Machine Vision

USE CASE Examples
- Object detection
- Image classification
- Facial recognition

APPLICATION
- Presence detector
- Security system
- Food recognition

facial recognition


Object detection

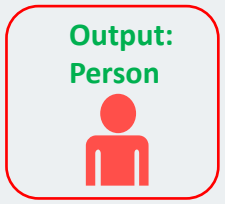


Image Classification

Speech AI

USE CASE Examples
- Keyword spotting
- Noise suppression
- Speech to text

APPLICATION
- Smart home controller
- Home appliance
- Interactive toys

Keyword spotting

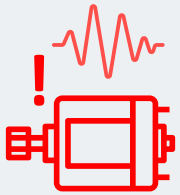
Noise suppression

Speech-to-text

Smart Sensing

USE CASE Examples
- Anomaly detection
- Sound recognition
- Gesture/Motion recognition

APPLICATION
- Industrial controller
- Smart home controller
- Home appliance


Anomaly detection

Sound recognition

Gesture recognition

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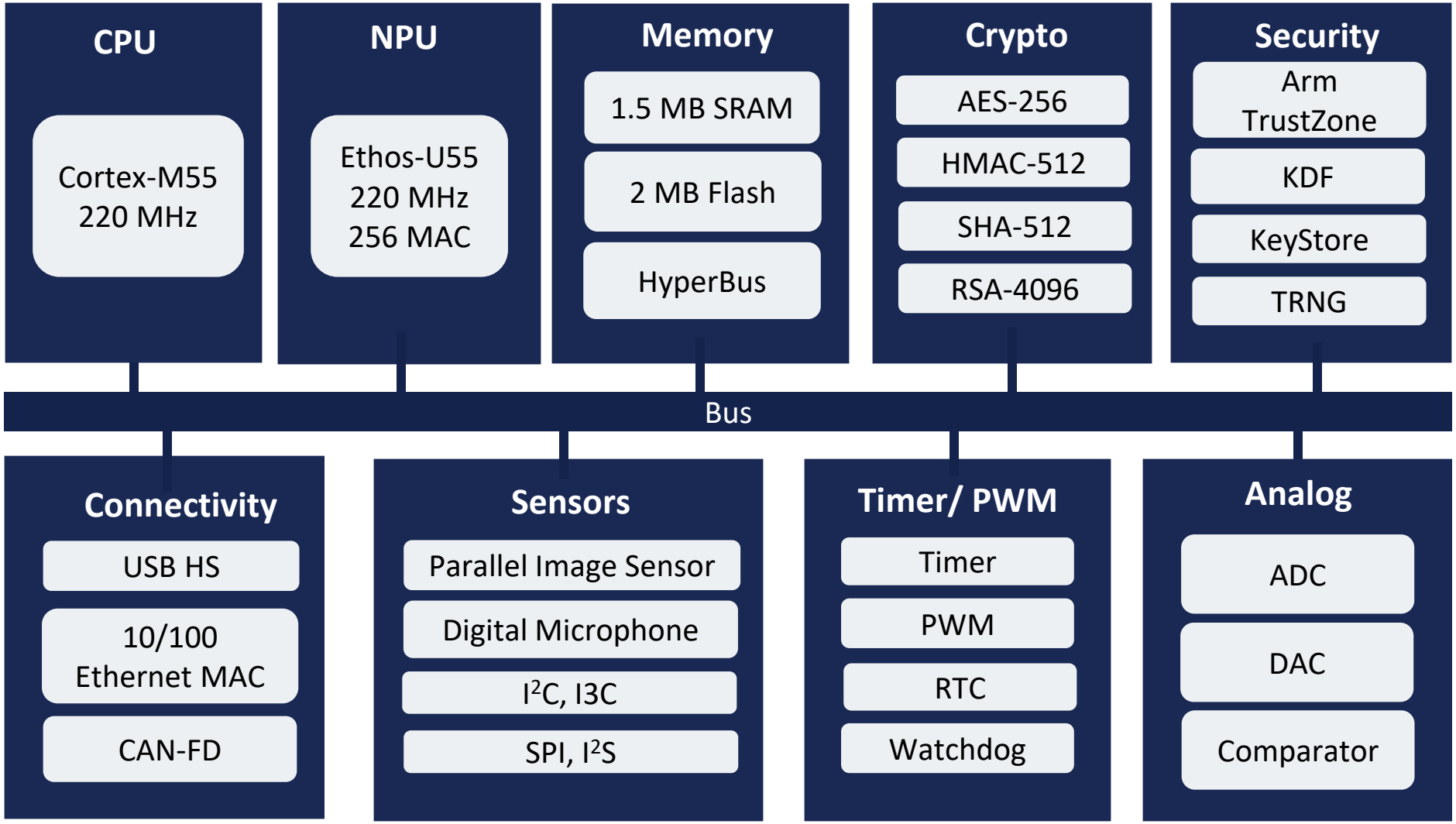
Introducing NuMicro® AI MCU - M55M1

A Full Function MCU

Security
Connectivity
Control
Analog

Extra AI Acceleration

256 MAC/cc
110 GOPS
CNN
RNN



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M55M1 MCU Key Features

M55M1's Key Features			
ML & DSP Acceleration	**Power management for energy-saving**	**IoT & Processor Security**	**Rich Peripherals**
■ **Cortex-M55 @ 220MHz** • Helium vector extension ■ **Ethos-U55 NPU @ 220MHz** • Dedicated for neural network • Support 256 x 8-bit MAC/cc. • Peak throughput = 110 GOP/s ■ Embedded memories • 1.5 MB SRAM • 2 MB embedded Flash memory ■ External memory interface • HyperRAM • HyperFlash • Octo-SPI flash • Support OTFC	■ Support 6 power modes • Active, NPD, LLPD, SPD, DPD, RTC w/ VBAT. • Functions supported in low power modes • DMA, I2C, SPI, Timer, UART, ADC, GPIO ■ Support threshold detection in camera, DMIC and sensor interfaces to save power consumption • Camera interface • Motion detection IP • DMIC (PDM) interface • Voice activity detection IP • Sensor interfaces • ADC/ACMP/I2C/SPI support threshold and timer monitoring.	■ TrustZone ■ Secure-boot ■ Crypto engines • AES-256 • SHA-512 • HMAC-512 • RSA-4096 • ECC ■ Key-store module ■ TRNG ■ PSA Level-2 Compliance	■ GPIO ■ UART ■ I2C ■ I3C ■ SPI ■ CAN-FD ■ PDMA ■ I2S ■ Camera - Parallel ■ ADC ■ ACMP ■ DAC ■ USB-2.0 ■ Ethernet MAC

Product Breif

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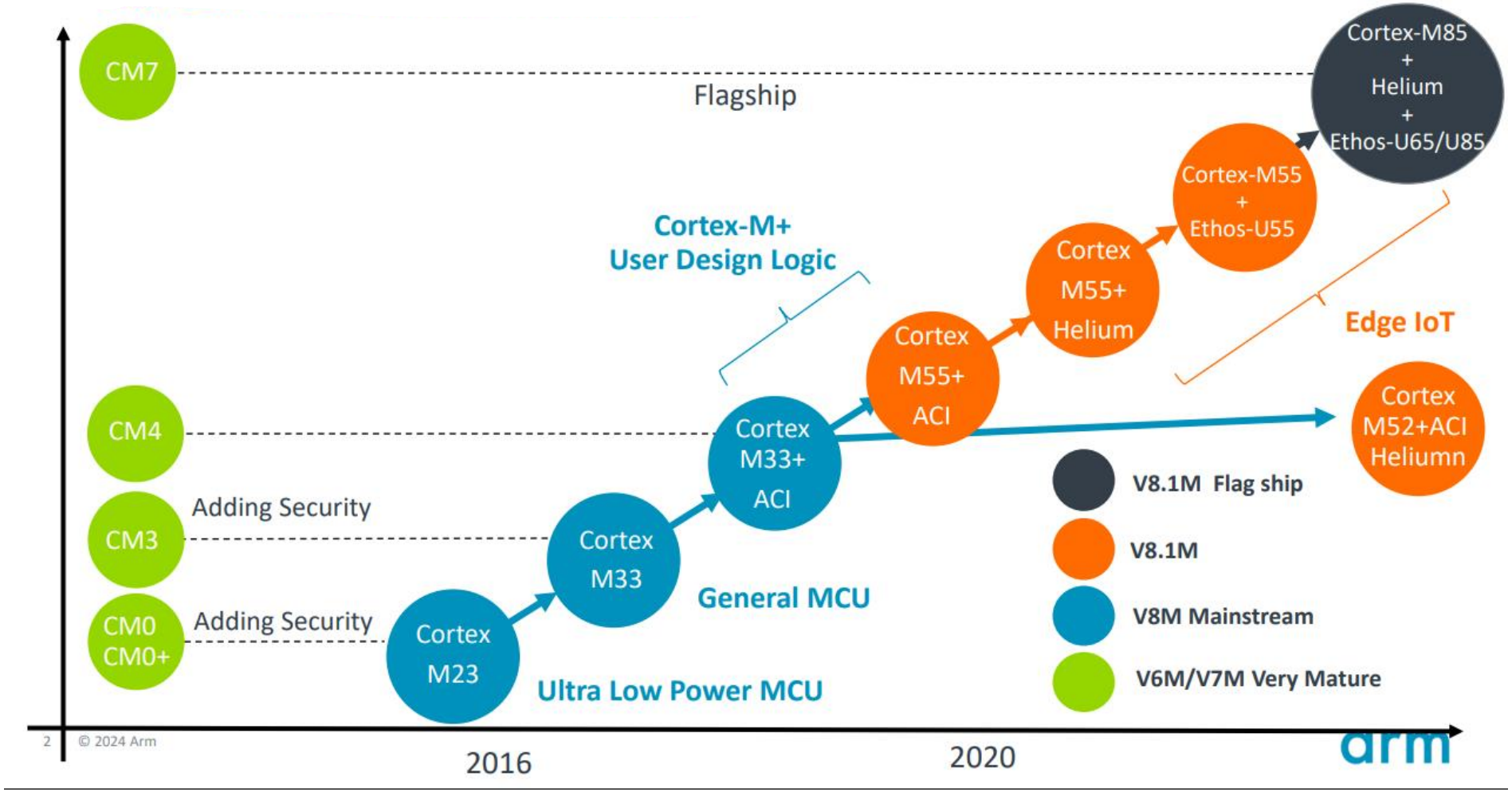
M55M1 NPU Key Features



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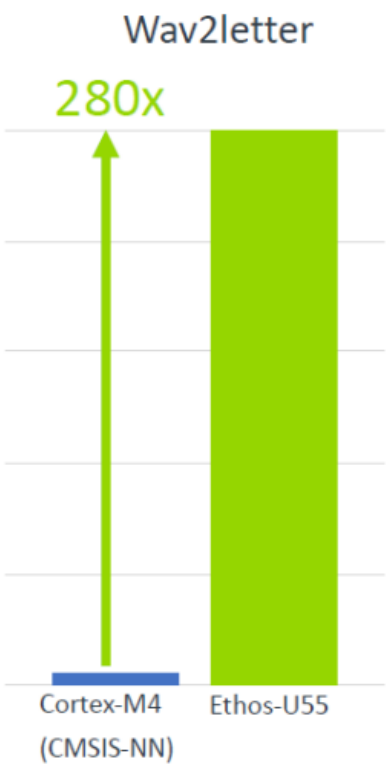
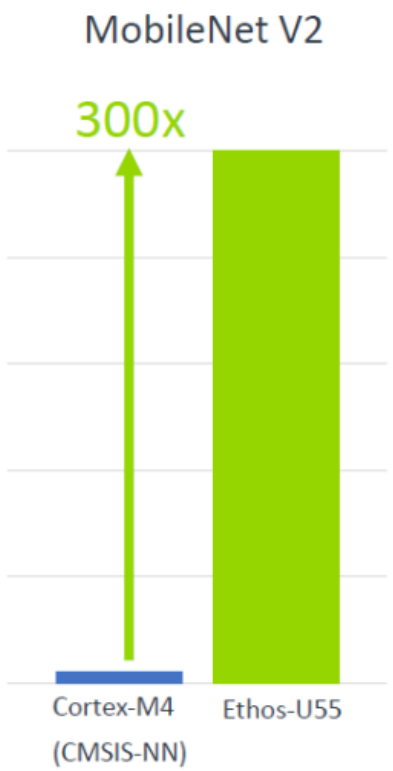
M55M1 NPU Key Features

Using ***256 MACs/Cycle*** configuration vs. Cortex-M4 using CMSIS-NN optimizations





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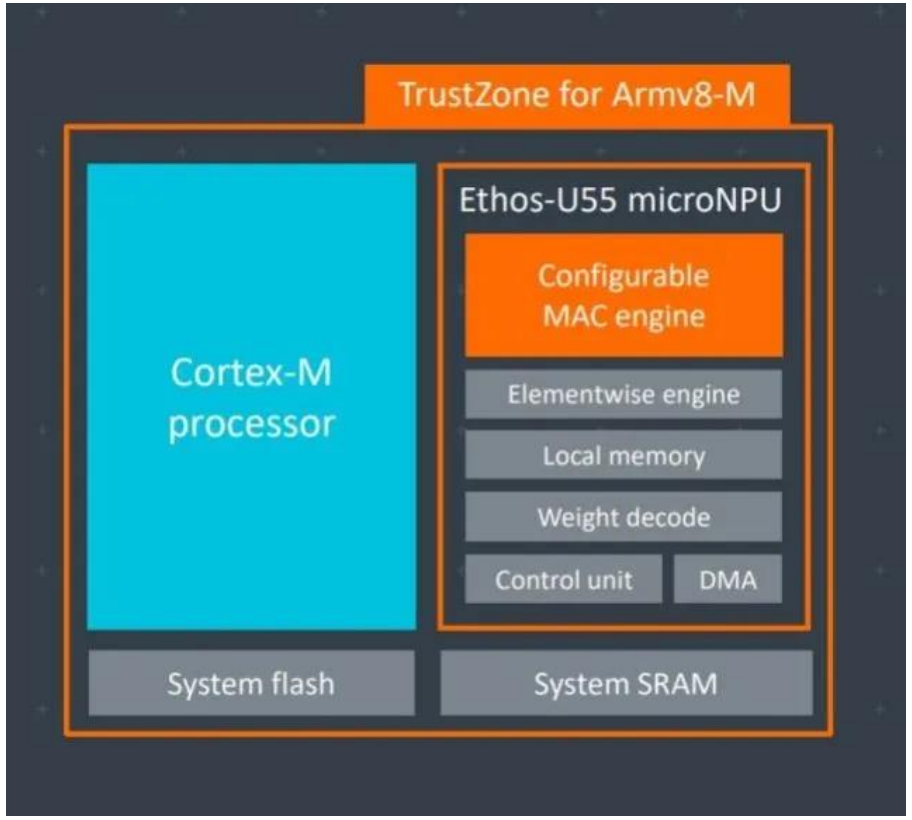
M55M1 NPU Key Features



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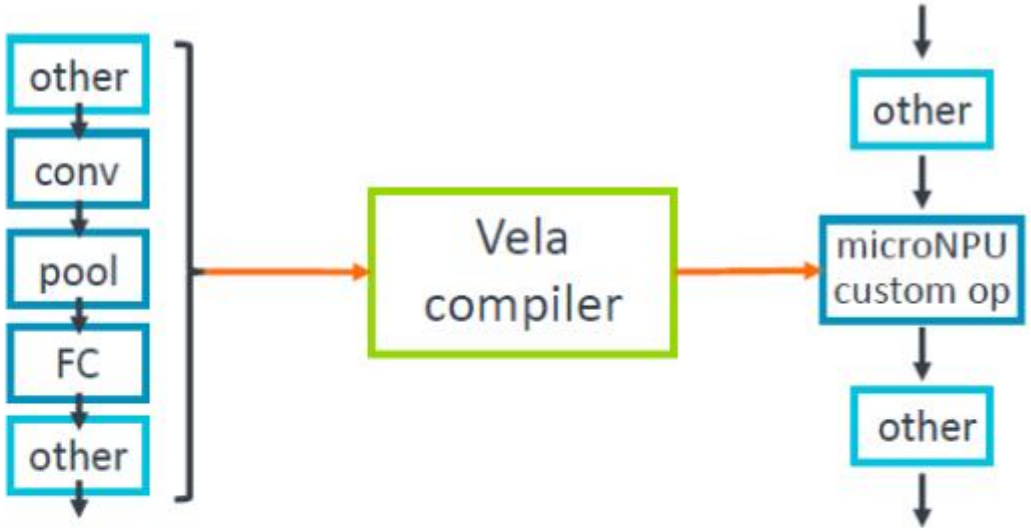
M55M1 NPU Vela Compiler

- Reads a tflite file, writes a modified tflite file
- Generates commands for microNPU
- Optimizes scheduling of subgraphs
- Loss-less compression of weights
- Reduces SRAM and Flash footprint
- Enabling networks previously not feasible in embedded systems!
- Open source



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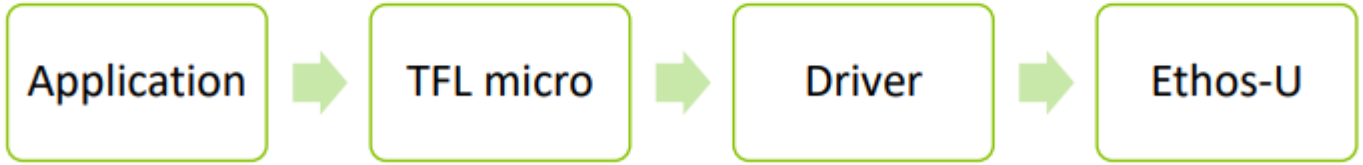
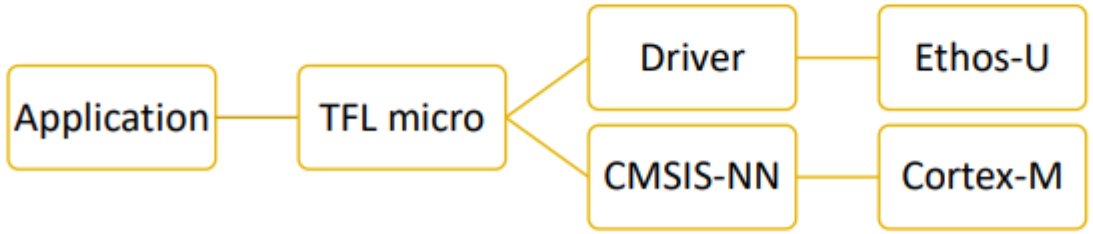
M55M1 NPU Execution Flow

- Ethos-U supports a fixed set of operators and can completely execute networks that map to that operator set. For ex:
  DS-CNN-L, ResNext 50, MobileNet v1, MobileNet v2, Inception v3, Inception v4, RNNoise, HAR_github, Deepspeech1, Wav2letter, … and many more



- For networks that cannot be executed on Ethos-U completely, the operators unsupported by Ethos-U fallback to the attached Cortex-M CPU
  - These are accelerated through CMSIS-NN library



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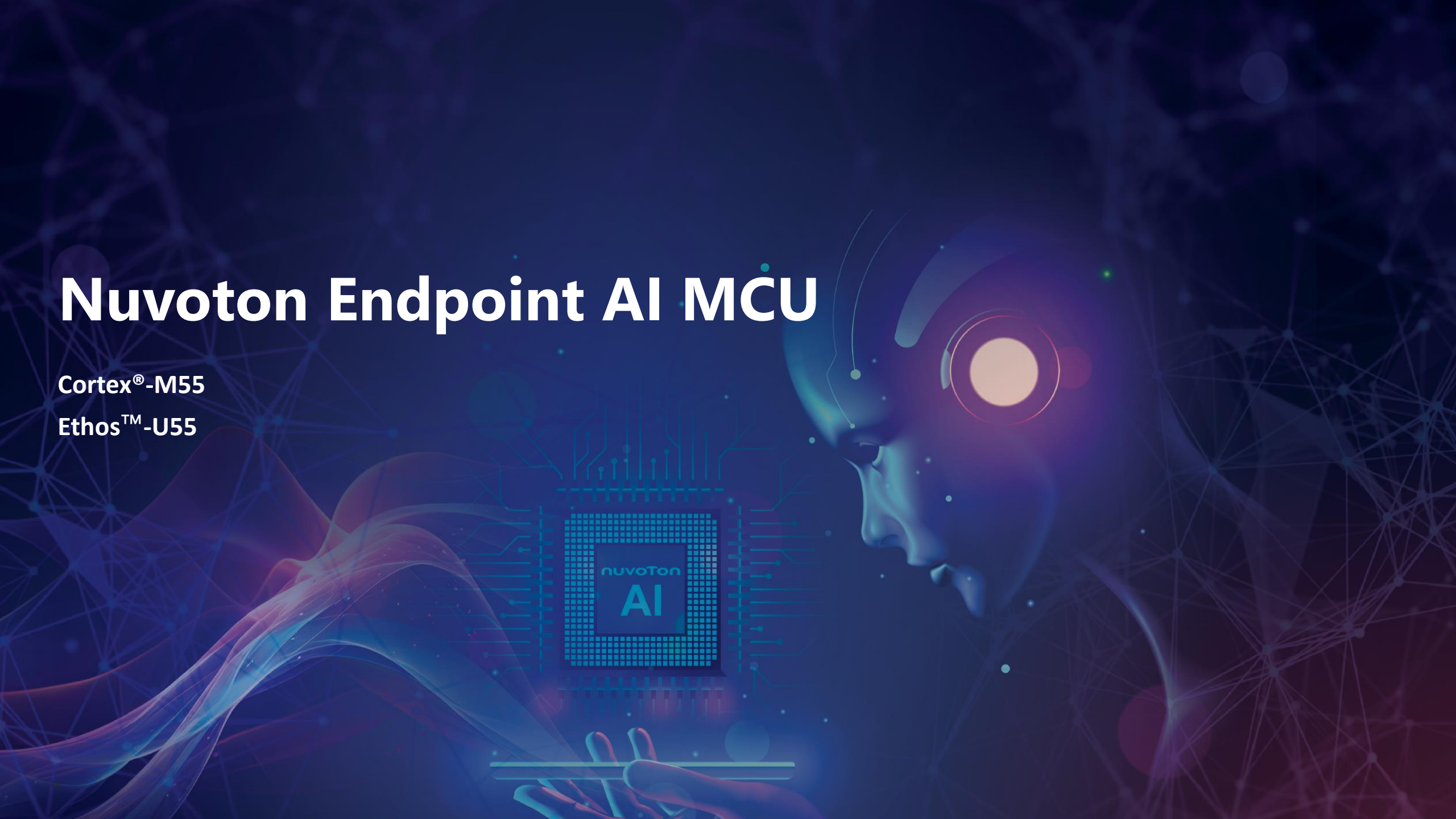
Nuvoton Endpoint AI MCU

Cortex®-M55

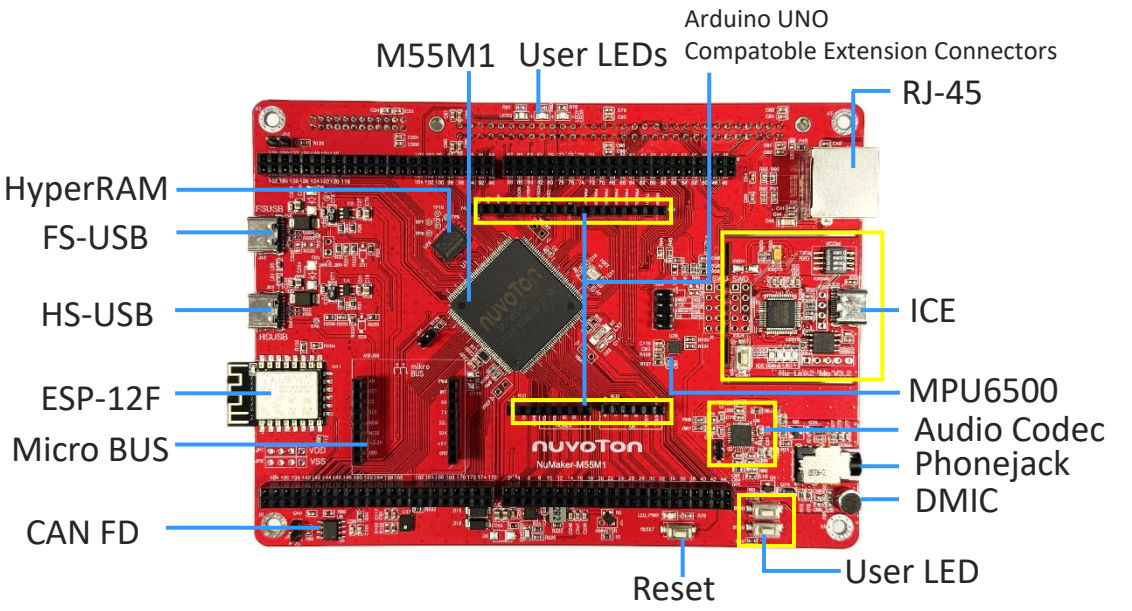
Ethos™-U55

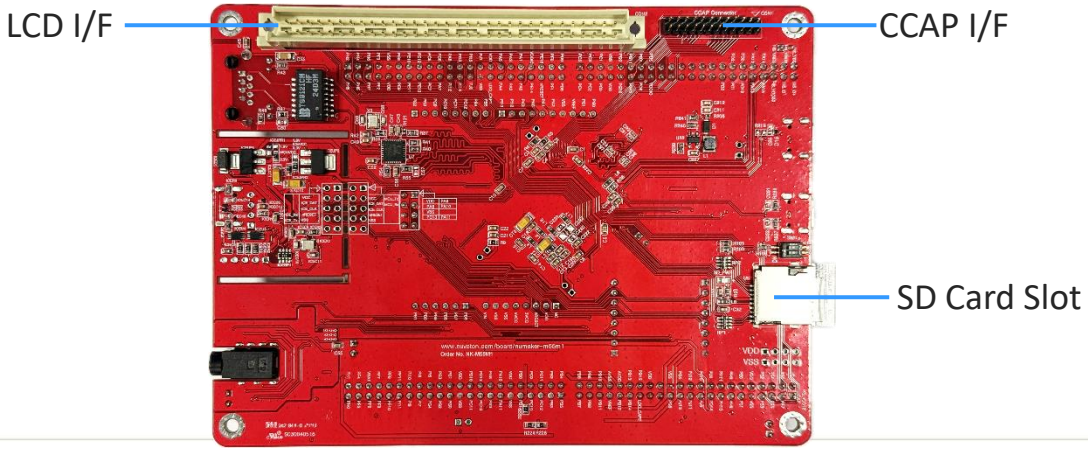


NuMaker-M55M1 EVB

- M55M1
  - Arm® Cortex®-M55 CPU, 220 MHz
  - Arm® Ethos®-U55 NPU, 220 MHz
  - Arm® M-profile Vector Extension (Helium)
- Memory
  - **SRAM 1.5MB, Flash 2MB**
- On-board components:
  - ***CMOS Sensor**
  - **Arduino UNO Extension Connectors**
  - **Nu-Link2-Me**
  - Gyroscope (MPU6500)
  - DMIC, MEMS microphone, Audio Codec
  - HyperRAM 8MB
  - User LEDs and user buttons
  - 10/100M ethernet PHY
  - FS-USB OTG and HS-USB OTG
  - Micro SD Card slot
  - CAN FD transceiver
  - Wifi module

User Manual





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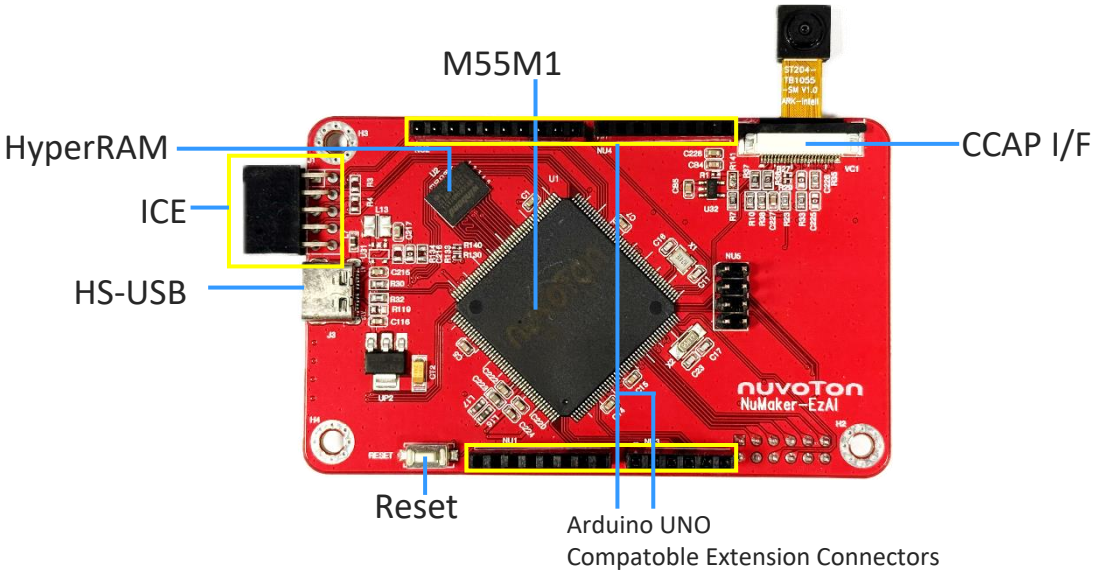
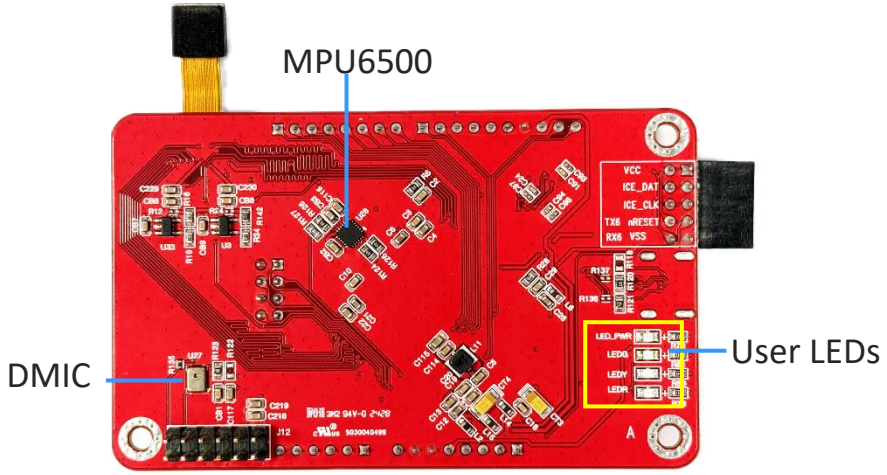
NuEzAI-M55M1 EVB

- M55M1 (NuEzAI)
  - Arm® Cortex®-M55 CPU, 220 MHz
  - Arm® Ethos®-U55 NPU, 220 MHz
  - Arm® M-profile Vector Extension (Helium)
- Memory
  - **SRAM 1.5MB, Flash 2MB**
- On-board components:
  - **CMOS Sensor**
  - **Arduino UNO Extension Connectors**
  - **Nu-Link2-Me I/F**
  - Gyroscope (MPU6500)
  - MEMS microphone
  - HyperRAM 8MB
  - User LEDs
  - HS-USB OTG





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Running Machine Learning on Microcontroller

Applications	Bring Your Own Model								
	KWS	Face Recog.	Img Classify	Obj Detect	Finger Gesture	Facial Landmark	Pose Landmark	Anomaly Detect.	Sound Recog.
ML Framework	TensorFlow								
ML Inference Runtime	TFLite Micro					TFLite			
System Software	CMSIS-NN ML Library	Ethos™-U55 Driver				Arm®-NN ML Library			
CPU/NPU	Cortex®-M4 CPU Cortex®-M55 CPU	Ethos™-U55 NPU				Cortex®-A35 CPU			

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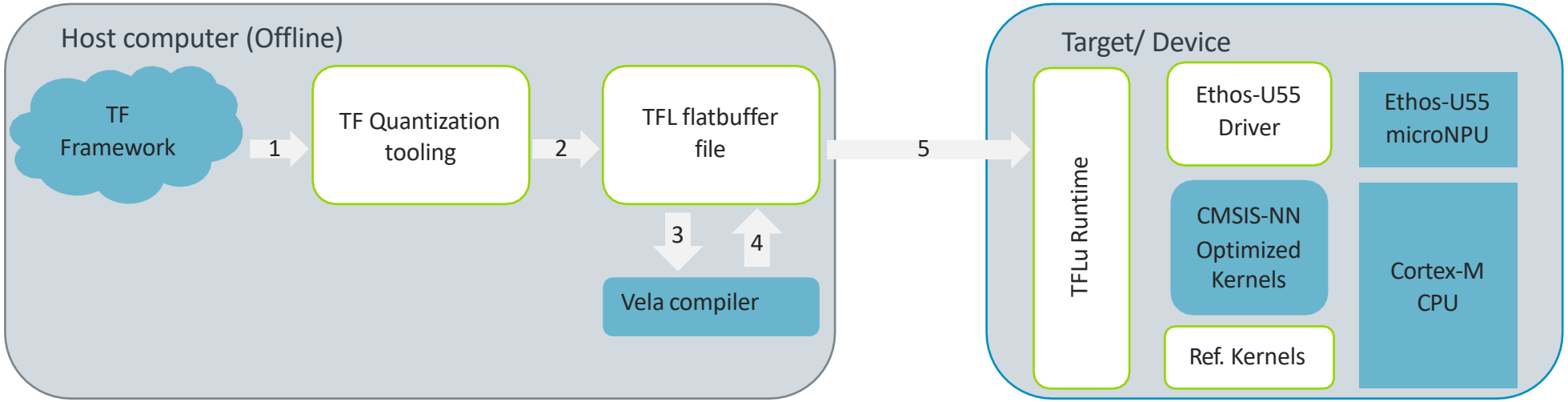
Model Deployment to Target



Host Computer (Offline)

- Train network in **TensorFlow**
- Quantize it to Int8 **TFL flatbuffer file** (.tflite file)
- **Vela compiler** identifies graphs to run on Ethos-U55
  - Optimizes, schedules and allocates these graphs
  - Lossless compression, reducing size of tflite file

Runtime executable file on device

- Accelerates kernels on **Ethos-U55** (Driver handles the communication)
- The **remaining layers are executed on Cortex-M**
- **Helium accelerates DSP** and ML tasks on Cortex-M with SIMD optimization

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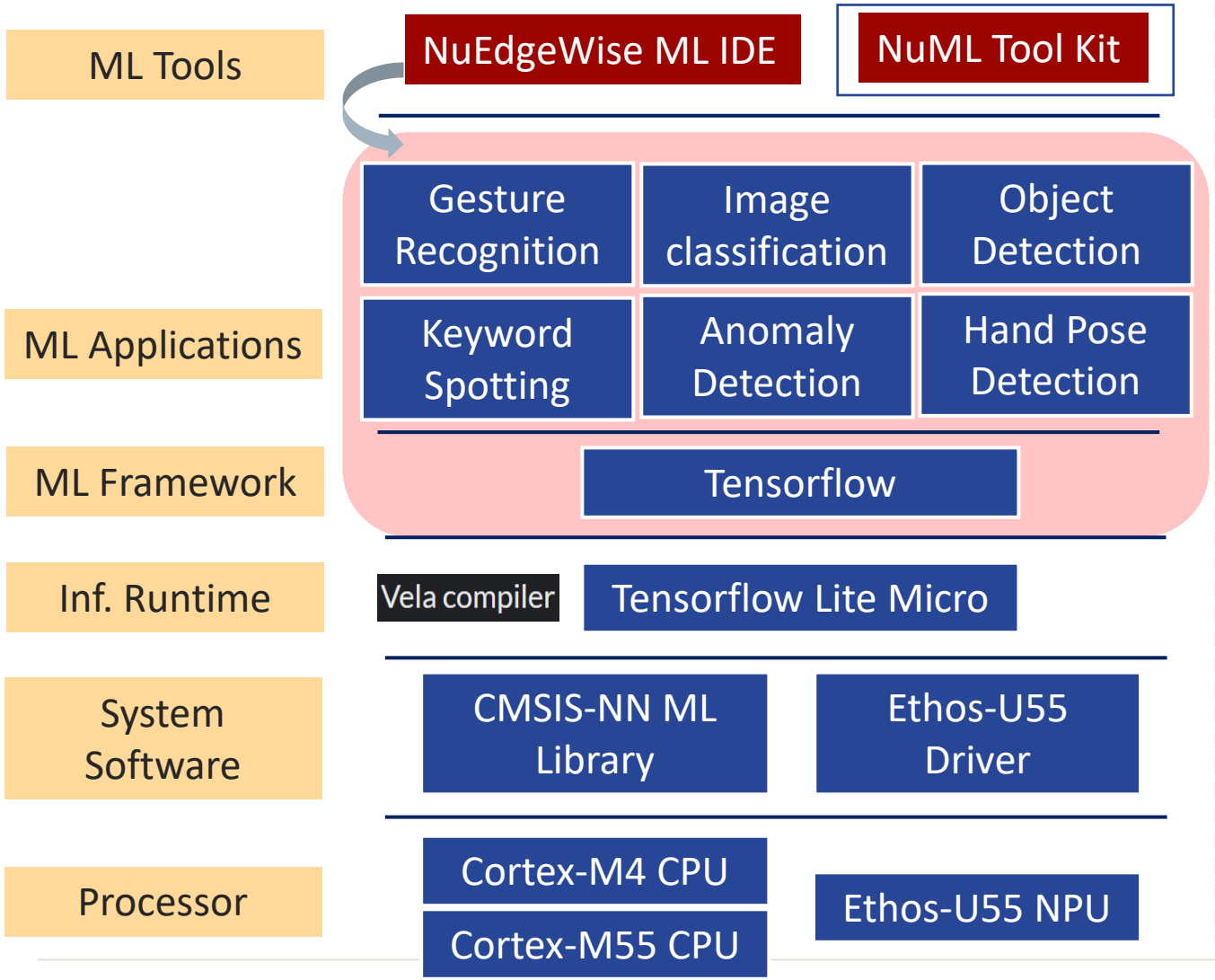
TFL: TensorFlow Lite

Enabling Valuable ML applications on MCU



Status and Work orientations

NuML Tool Kit (Key Features):

1.Model Preparation:

1. Users prepare a TensorFlow model.
2. Convert the TensorFlow model to TensorFlow Lite (.tflite) format.

2.Seamless Integration:

1. Upload the .tflite model to the tool.
2. The tool handles automatic deployment without additional configurations.

3.Deployment Process:

1. Efficient and fast model deployment in just a few steps.
2. Compatible with various AI applications on the M55M1 board.

Diagram :

1. Model preparation (TensorFlow) → 2. Convert to .tflite → 3. Upload to tool → 4. Automated deploy to M55M1.

Vela Official Web

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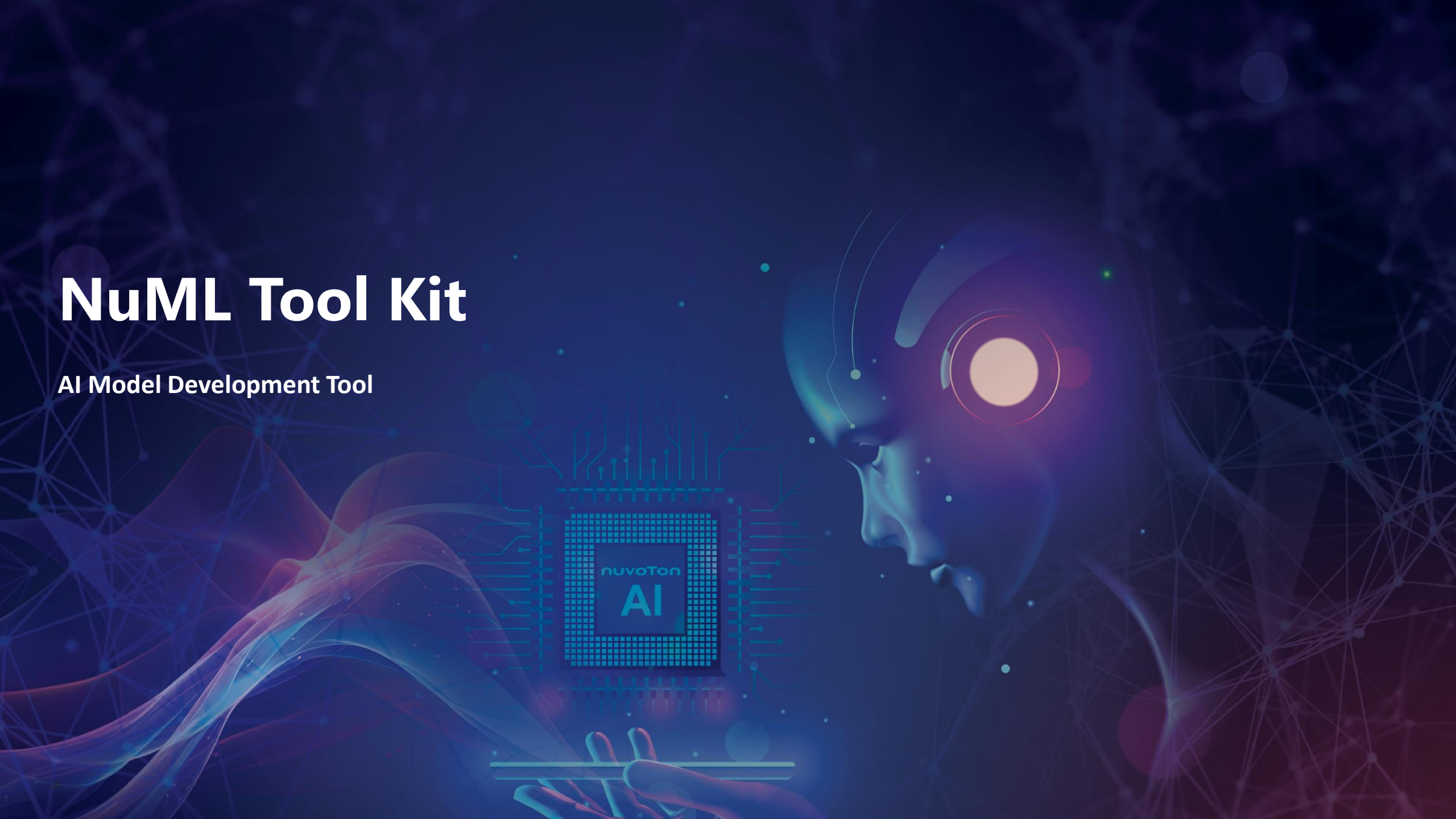
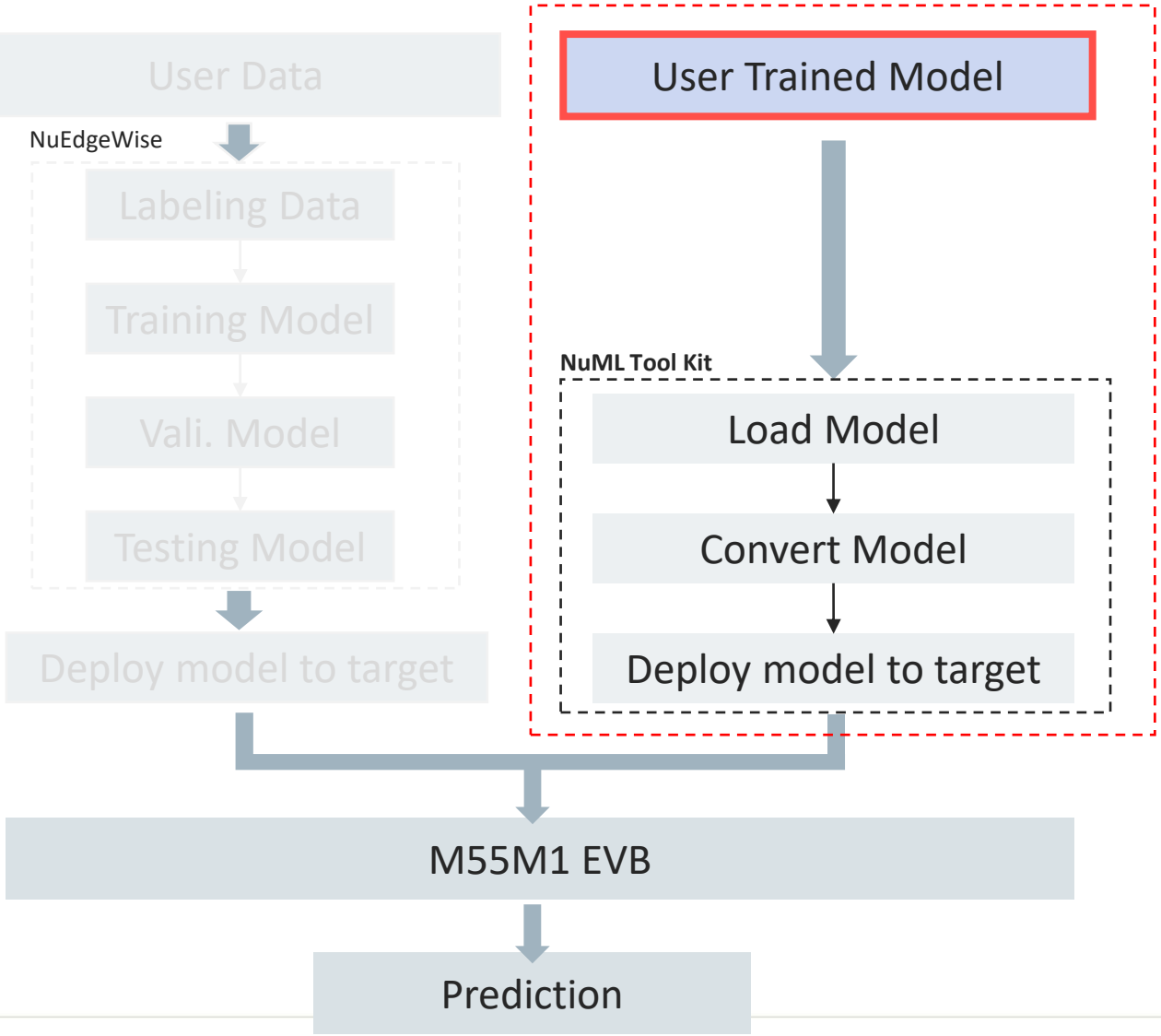
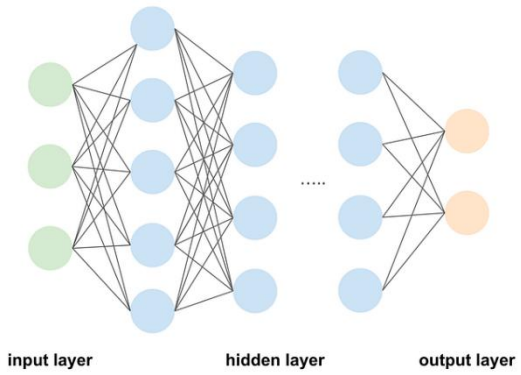
NuML Tool Kit

AI Model Development Tool



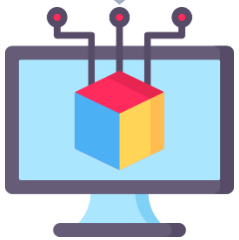
NuML Toolkit Block Diagram



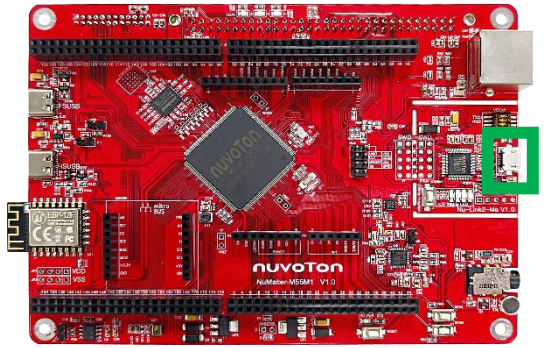


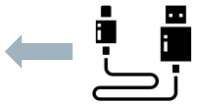
TensorFlow

TFLite Model



NuML Tool Kit





M55M1 EVB

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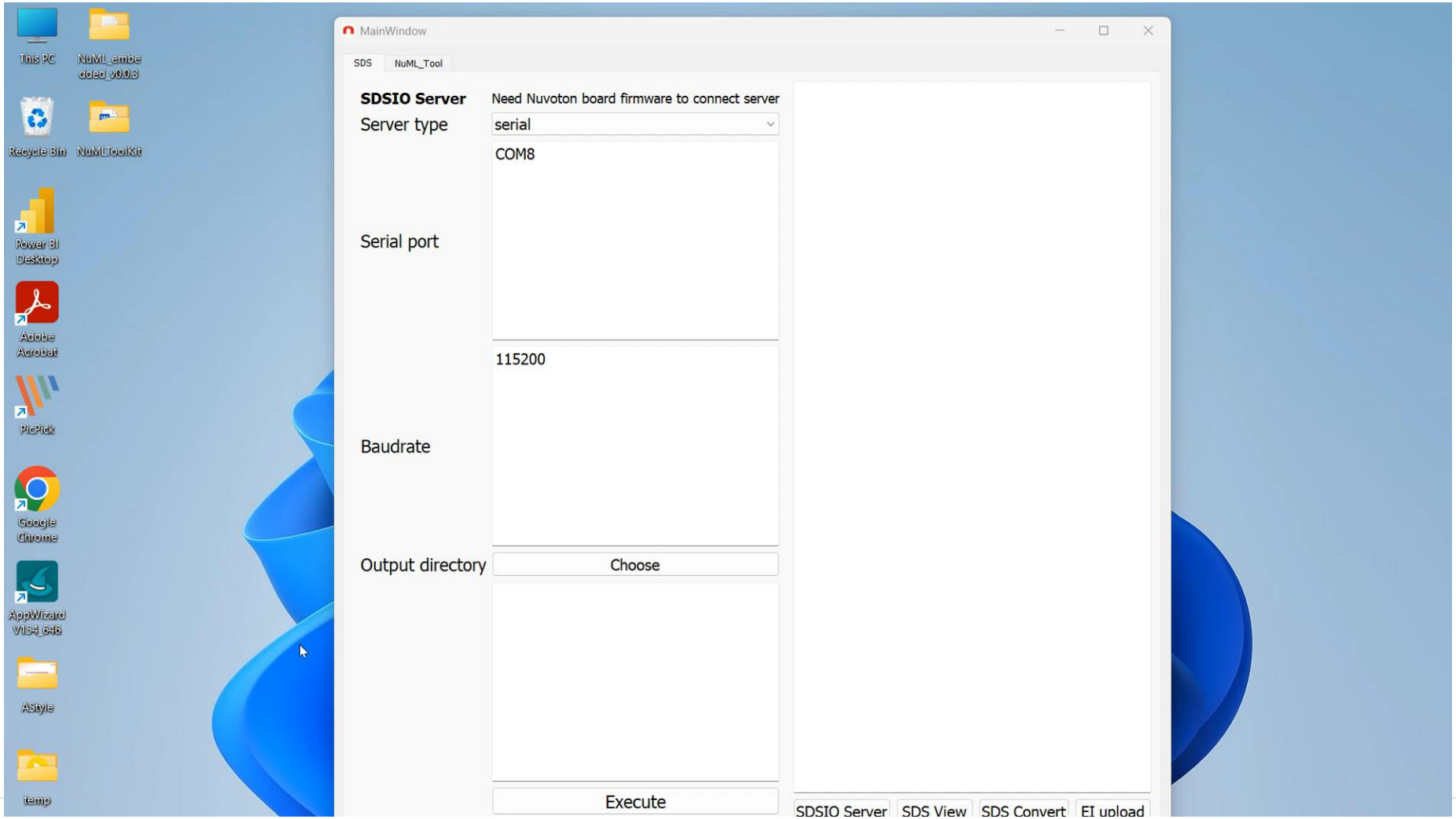
NuML Toolkit Block Diagram

Download Link



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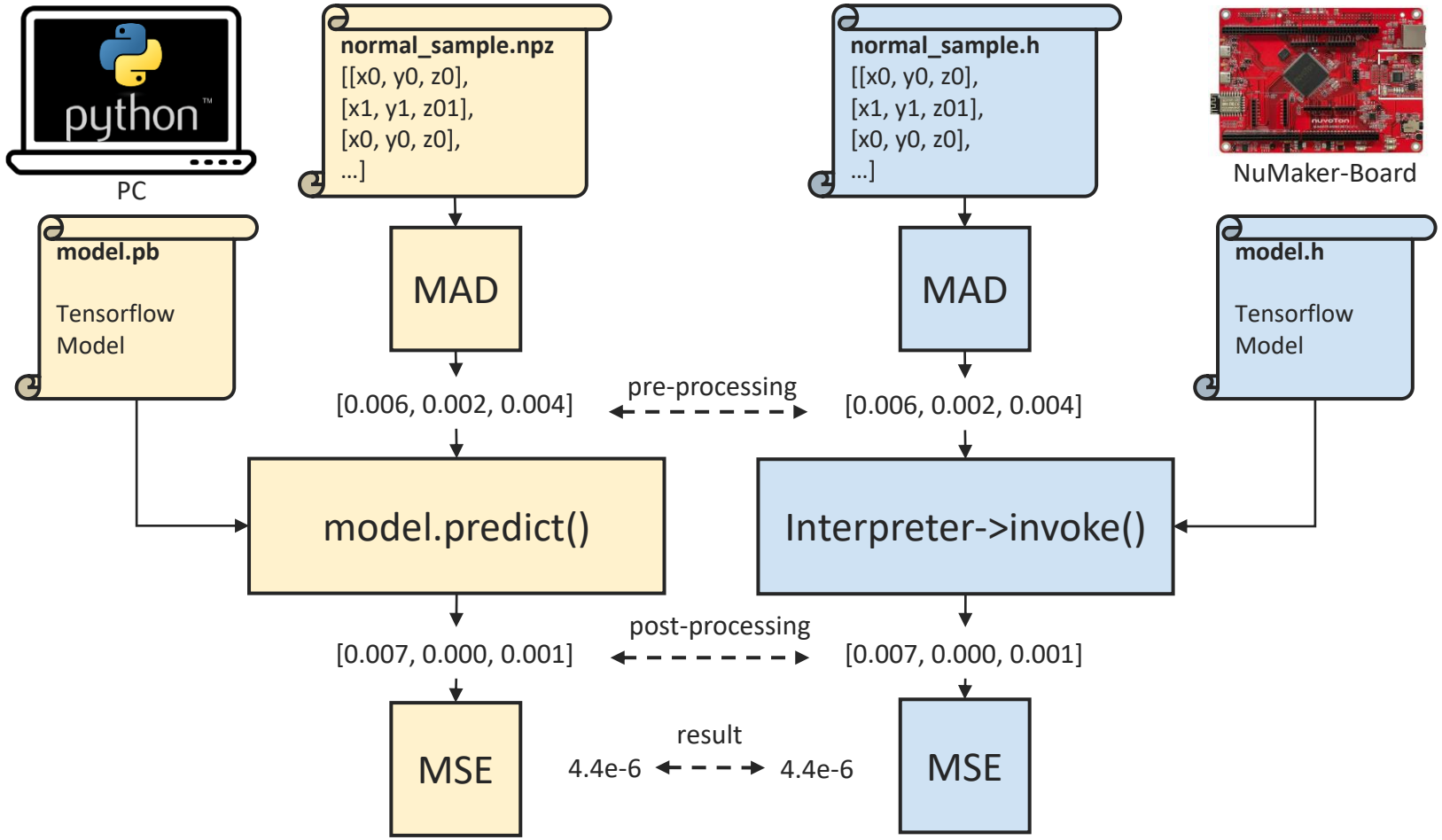
Tensorflow lite micro Flow



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AI Development Environment

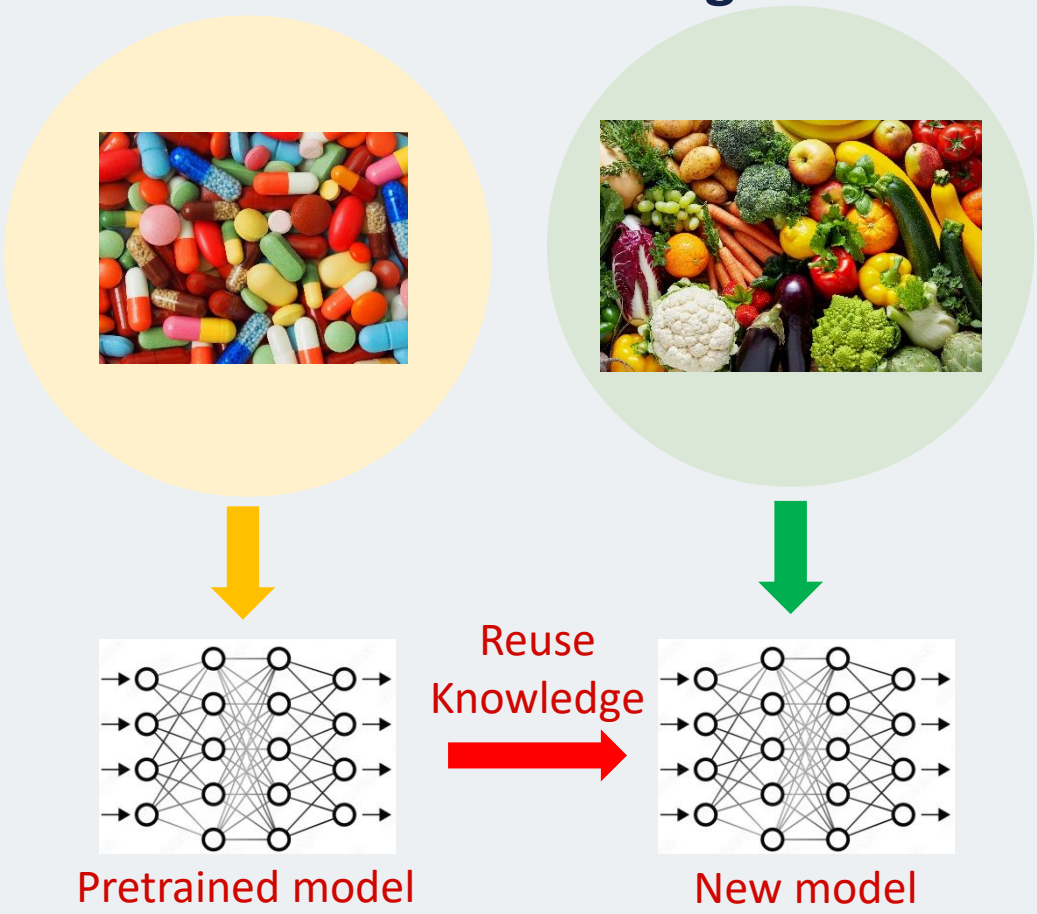
NuEdgeWise

NuEdgeWise Machine Learning Model Zoo

NuEdgeWise ML Model Zoo

KWS	Face Recog.	Img Classify
Obj Detect	Finger Gesture	Facial Landmark
Pose Landmark	Anomaly Detect.	Sound Recog.

Transfer Learning



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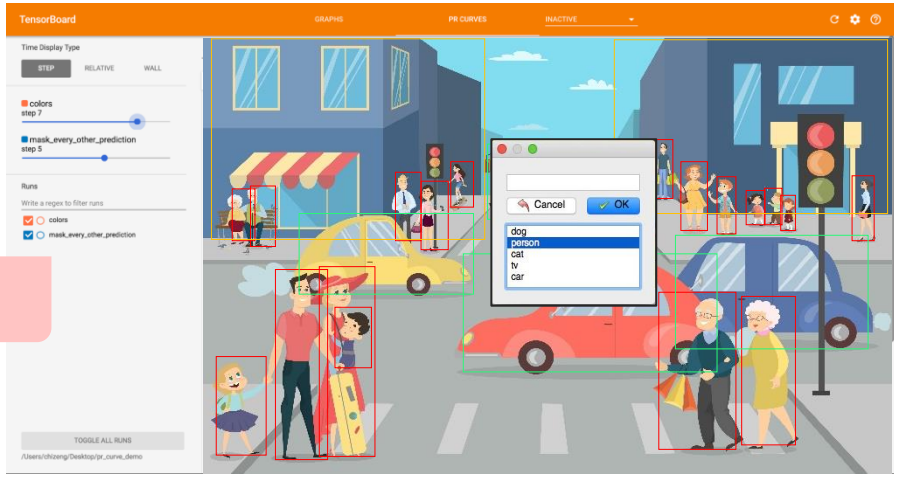
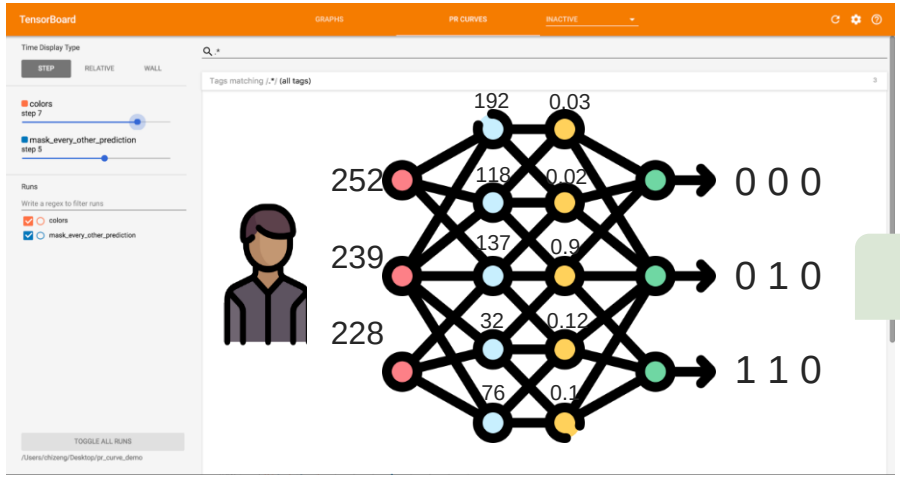
NuEdgeWise IDE – Building Your Own Model

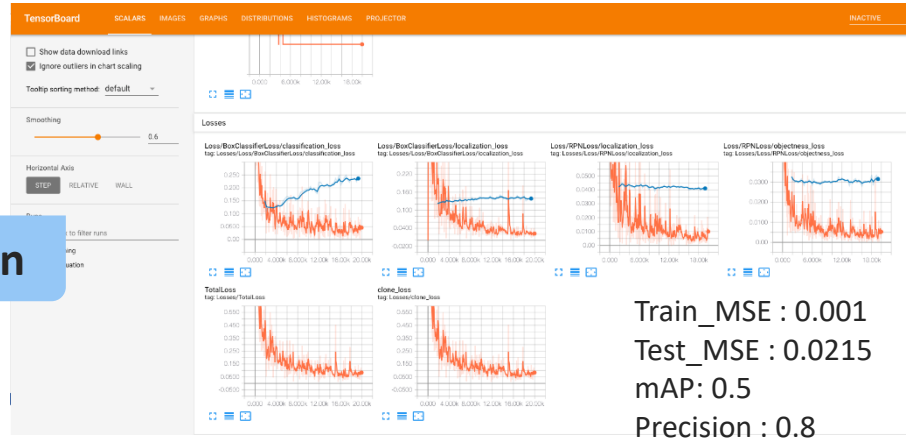
Labeling



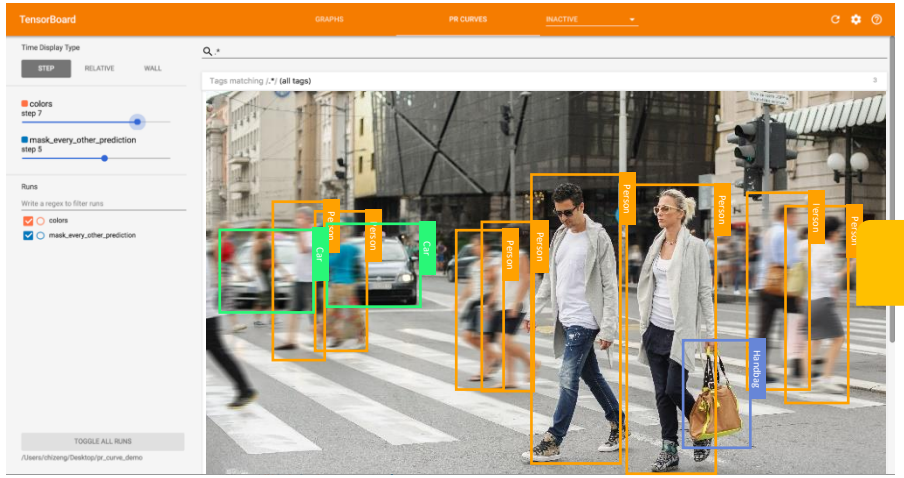


Training

Validation



Train_MSE : 0.001
Test_MSE : 0.0215
mAP: 0.5
Precision : 0.8



Testing

https://github.com/OpenNuvoton/NuEdgeWise
Quick Start Guide

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NuMicro ML and M55M1 Development Resources

- NuEdgeWise ML Model Zoo
  - https://github.com/OpenNuvoton/NuEdgeWise
- M55M1 Board Support Package (BSP)
  - https://github.com/OpenNuvoton/M55M1BSP
- M55M1 EVB
  - NuMaker-M55M1
  - NuEzAI-M55M1
    - https://www.nuvoton.com/ai/product/
- Suggestion Information
  - Link

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Training Videos

Subject		Introduction	Sample code tutorial
Beginner Course		https://www.youtube.com/watch?v=u74iwBIC3I0	
Level 1 Workshop	Timer	https://www.youtube.com/watch?v=c9OM2sKagOE	https://www.youtube.com/watch?v=hpH3Le7TBbk
	WDT/WWDT	https://www.youtube.com/watch?v=KZtL97XIxj0	https://www.youtube.com/watch?v=rrjStxOHkLI
	I2C	https://www.youtube.com/watch?v=1Wfdcvs-X_Y	https://www.youtube.com/watch?v=-tN67Gh3iRQ
	UART	https://www.youtube.com/watch?v=bbNCv1k1apU	https://www.youtube.com/watch?v=tLdfqDK-fPA
	SPI	https://www.youtube.com/watch?v=fMtb4yxBBN4	https://www.youtube.com/watch?v=hGEihw0OA5A
	ADC	https://www.youtube.com/watch?v=_JiEubRen48	https://www.youtube.com/watch?v=alqyHk3-We8
	PWM	https://www.youtube.com/watch?v=VMu4nOTP2R4	https://www.youtube.com/watch?v=1qjwDOcVT_M
	GPIO	https://www.youtube.com/watch?v=W8fLTAeJ2fA	https://www.youtube.com/watch?v=FtfGWg0z7Es
	EPWM	https://www.youtube.com/watch?v=VVa_yuuguUo	n/a
	USB	https://www.youtube.com/watch?v=JJeP0vMcGo8	n/a
Debugger and Programmer		https://youtube.com/playlist?list=PL0M6qIz4YrQ2-DaAk0swXcH7_mvohtNcY	
NuTool - PinConfigure		https://www.youtube.com/watch?v=nkWMOGWF-Pk	
NuTool - PinView		https://youtu.be/AhTA_roYfMQ?si=nDN3fYfxmi3Yebe-	
NuTool - LCDView		https://youtu.be/TXKK-aRGHuU?si=wn_XLm7BSC-_WPaN	
(optional) M467 with SEGGER emwin AppWizard		https://youtu.be/phonD1kuJxU?si=TJvge7iiADWihY50	
(optional) Capacitive touch intro		https://youtu.be/TX38Ob838SI?si=fIh62191ul91tflV	
(optional) Touch key tuning utility		https://youtu.be/TL7-l3tjvpY?si=jxI3o251yeLsBZ4f	

Level 1 Workshop

HMI emWin

Touch Key

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NuMicro® Training Videos

Official Youtube

https://www.youtube.com/@NuMicro

Beginner

Beginner Course

https://reurl.cc/eGW9zL

Debugger and Programmer

https://reurl.cc/oVZ9yV

Level 1 Workshop

https://reurl.cc/Q5W7e5

Intermediate

NuTool-PinConfigure

https://reurl.cc/kMqVyG

NuTool-PinView

https://reurl.cc/nqZnmn

NuTool-LCDView

https://reurl.cc/6jLyKk

Advanced

M467 with SEGGER emwin AppWizard

https://reurl.cc/p9ZmYr

Capacitive touch intro

https://reurl.cc/b3Gzmr

Touch key tuning utility

https://reurl.cc/961Zbv

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NuMicro® Digital Platform

Official Website

www.nuvoton.com

Online Store

direct.nuvoton.com

nuvoton.tmall.com

www.techdesign.com/market/nuvoton

Forum

forum.nuvoton.com

www.nuvoton-MCU.com

bbs.21ic.com/iclist-187-1.html

Social Media

www.facebook.com/NuvotonNuMicro

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Source Code Download

github.com/OpenNuvoton

gitlab.com/OpenNuvoton

gitee.com/OpenNuvoton

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Reference Video

Type	Topic	Tech Blog	AI Mini Site	Video
Product/Tool	NuEzAI-M55M1	NuEzAI-M55M1 Development Board: A Powerful Tool for Simplifying AI Development	AI Product	EN / TC / SC / KR / JP
Application	Medicine Classifier	Medicine Classifier	AI Discovery	EN / TC / SC / KR / JP
Application	Gesture Recognition	Nuvoton Gesture Recognition: A High-Performance Application For MCU AI	AI Discovery	EN / TC / SC / KR / JP
Application	Body Pose Landmark	Implementing Pose Landmark Detection using the NuMicro® M55M1 ML MCU	AI Discovery	EN / TC / SC / KR / JP
Application	Facial Landmark	Utilizing the NuMicro® M55M1 ML MCU for Facial Landmark Detection Applications	AI Discovery	EN / TC / SC / KR / JP
Application	Voice Control	Voice Control Scope	AI Discovery	TC
Application	Auto Plate Billing	Auto Plate Billing	AI Discovery	TC
Application	Warehouse	Warehouse Management System	AI Discovery	TC
Application	People Counting	Meeting Room People Counting	AI Discovery	EN / TC / SC / KR / JP

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Joy of innovation

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谢谢

謝謝

Děkuji

Bedankt

Thank you

Kiitos

Merci

Danke

Grazie

ありがとう

감사합니다

Dziękujemy

Obrigado

Спасибо

Gracias

Teşekkür ederim

Cảm ơn